Project Work Report

Nassau Candy Distributor – Factory Reallocation & Shipping Optimization

1. Project Overview

In this project, I analyzed the Nassau Candy Distributor dataset to understand sales performance, profitability, regional distribution, product performance, and shipping operations. The objective was to use data analytics to identify potential areas for operational improvement and support data-driven decision-making.

2. Dataset Description

The dataset contains **10,194 rows and 18 columns**. It includes order information, order and ship dates, shipping modes, customer locations, product details, sales, units, gross profit, and cost.

3. Work Performed

3.1 Data Understanding

- Inspected the dataset structure, column names, and available business information.
- Identified important fields required for sales, profitability, regional, and shipping analysis.
- Reviewed order dates and ship dates for lead-time calculations.

3.2 SQL Analysis

- Calculated key business metrics, including total sales, gross profit, units sold, unique orders, and unique customers.
- Analyzed sales performance by region and division.
- Examined product-level sales and profitability.
- Compared shipping modes using order-to-shipment lead time.

- Investigated monthly sales trends, city-level performance, and records with negative gross profit.

3.3 KPI Identification

Identified key metrics for the project, including:

- Total Sales
- Total Gross Profit
- Total Units Sold
- Total Orders
- Profit Margin
- Order-to-Shipment Lead Time

Additional proposed optimization KPIs include Lead Time Reduction, Profit Impact Stability, Scenario Confidence Score, and Recommendation Coverage.

3.4 Power BI Dashboard Planning

Designed a dashboard structure to present the analysis through KPI cards, regional sales comparisons, monthly sales trends, top-performing products, shipping lead-time analysis, and interactive filters for region, division, shipping mode, and order date.

3.5 Optimization Approach

Outlined a proposed approach for comparing baseline performance with alternative shipping and resource-allocation scenarios. These scenarios would need additional assumptions and validation before estimating actual operational improvements.

4. Key Findings

The initial analysis identified the following metrics and observations:

- Total sales: approximately **$141,783.63**.
- Total gross profit: approximately **$93,442.80**.
- Total units sold: **38,654**.
- Unique orders: **8,549**.
- Unique customers: **5,044**.
- The Pacific region recorded approximately **$46,301.53** in sales.

The analysis also examined product profitability, monthly sales patterns, shipping modes, and negative-profit records. Shipping lead-time results require additional date validation because unusually high values were observed.

5. Tools and Technologies

- **PostgreSQL:** SQL queries, aggregations, filtering, and business analysis.
- **Microsoft Power BI:** KPI reporting and dashboard visualization.
- **Microsoft Excel / CSV:** Dataset inspection and data handling.

6. Learning Outcomes

Through this project, I practiced translating business questions into SQL queries, calculating performance indicators, analyzing sales and profitability, examining shipping data, and planning an interactive business dashboard. I also learned the importance of validating dates, checking unusual values, and distinguishing observed results from projected improvements.

7. Limitations and Future Scope

The current dataset provides order and shipment dates but does not establish the final delivery date or actual factory capacity constraints. Future work can include validated factory allocation data, shipping cost analysis, scenario comparison, and testing recommendations against historical performance.

8. Conclusion

This project demonstrates how SQL and Power BI can be used to analyze distributor operations and communicate important business metrics. The analysis provides a foundation for further investigation into shipping efficiency, profitability, and resource allocation. Any claimed reduction in lead time or improvement in profit should be reported only after the proposed scenarios have been calculated and validated.